

Reviewer guide: how to review orbit_demo

This guide takes a reviewer through the package in a fixed order, says what to check at each step, and ends with a feedback template. The package [README.md](#) has the exact revision, folder map and status legend; this guide does not repeat them.

What is being reviewed. orbit_demo is a 4-lane INT8 MAC digital demonstrator with duplicated storage and a mismatch fault-stop, plus a thermal FSM with TMR state. It is implemented as a core-only standard-cell block on SkyWater sky130hd. The reviewed layout is the ORFS run sep at 7.2 ns with copy-separation fences. Timing was optimised at TT only and fails setup at ss_100C_1v60 (K-01). It has not been fabricated, has no pad ring or package, and has had no radiation testing (K-05, K-08).

Out of scope. The unbuilt 16-tile concept chip in the brief, the reference layouts (base at 7.0 ns, IHP SG13G2), and the presentation material in site/, viz/ and docs/research/. The review notes list where the presentation material overstates the evidence (D-62 to D-84); question 14 asks only whether it should be corrected or withdrawn.

What you need

Purpose	Needed
Reading (steps 1-4)	A PDF viewer, or GitHub's Markdown view. Each section's text is a .md file, and the main ones also have a .pdf. Step 4 also needs sha256sum
Inspecting the layout	KLayout (the package used 0.30.12) to open 06_physical_design/layout_db/6_final.gds.gz . The sky130 layer-properties file is not in the package, so KLayout uses default colours. The pre-rendered views in 06_physical_design/klayout_views/ use the ORFS sky130hd.lyp
Re-running checks (step 5, optional)	A checkout of the repository at the package commit and the OSS CAD Suite. The layout rows also need Docker with the ORFS image and the outputs of a make pd-sep run. See 04_digital_design/README.md section 6

Time plan

Step	Activity	Time
1	Orientation	20 min
2	Known issues and the reviewer questions	30 min
3	Area-by-area review (seven areas)	1.5-3 h
4	Revision and consistency check	15 min
5	Reproduce selected results (optional)	15 min to 1 h
6	Write feedback	20 min

Total without step 5: about 3-4.5 h.

Step 1. Orientation (20 min)

1. [README.md](#): read the revision table, the status legend and the verification status overview.
2. [01_overview/design_overview.pdf](#): read the scope split (built demonstrator vs concept-chip TARGETs), then the requirements table. Note that the demonstrator has **no** specified clock target, supply specification, power budget or PVT range (all MISSING, G-01). Every timing or power number in the package is therefore a result without a requirement to compare against.
3. [02_block_diagram/orbit_demo_top_level.pdf](#): keep this open for the rest of the review. The detailed diagram is [orbit_demo_block_diagram.pdf](#).

Step 2. Known issues and questions (30 min)

Read [09_review_notes/review_notes.pdf](#) sections 1-3.

Section 1, known issues K-01 to K-16. The five High items:

- K-01: setup fails at ss_100C_1v60.

- K-02: single points of failure.
- K-03: common-mode faults are not detected, and several are not injected.
- K-04: the DRC deck runs BEOL and off-grid checks only (FEOL off), and LVS is not independent.
- K-05: no pad ring or package.

For each, decide whether it blocks the next step, and record any you disagree with.

Section 2, trade-offs T-01 to T-07. Check that each trade-off is justified by its cited evidence.

Section 3, questions 1-14. These are the decisions the design owner is asking you to make. Answer them in your feedback (step 6). The area checklists below point to the questions that each area informs.

Step 3. Area-by-area review

Work through the areas in order. Each checklist item names the document and, where relevant, the review-notes item it relates to.

3.1 Architecture and RTL

Documents: [04_digital_design/README.md](#) sections 3-5; RTL in [04_digital_design/source/rtl/](#); [source/docs/SPEC.md](#); schematics in [03_schematics/](#).

- [] Compare the RTL with SPEC sections 3-6: the MAC equation and wrap-around, the in_ready/out_valid equations, the duplicate-and-compare and sticky fault_q, the TMR vote and feedback repair, and the throttle/stop/recover thresholds. Known differences: phase wording (K-16, D-02), a presented result withdrawn by clear_fault or rst_n (D-03), and the combinational clear_fault -> in_ready path (D-04).
- [] Check the fault model against SPEC section 8 and K-02/K-03. One instance feeds both copies: the multiplier and prod32, the comparators and the mismatch OR-tree, the lane control (in_fire, clr, in_first, in_last), the thermal voter and the nxt cone, the clock, the reset and the inputs. phase, fault_q and out_valid_q are unprotected. SPEC section 8 lists only part of this (D-01). Questions 3-6.
- [] Reset and clear: rst_n is synchronous with no on-chip synchronizer. clear_fault zeroes lane storage but not the thermal state. Question 8 covers the host re-send protocol.
- [] Clocking: single domain, no CDC, no clock gating in the RTL.
- [] Schematics: [03_schematics/README.md](#) explains why transistor sizes and bias currents are N/A (standard cells only). The gate-level sheets come from the synthesized netlist, before placement; 03 lists the differences from the routed netlist.
- [] ECC: [rtl/ecc](#) is standalone and not in the layout (K-12). Decide whether it should be reviewed now.

3.2 Functional verification

Documents: [05_simulation/simulation_report.pdf](#); formal and fault sections of [07_verification/verification_summary.pdf](#).

- [] Test inventory S1-S14: check that the stimulus, pass criterion and checked properties cover each SPEC requirement. Use the requirements table in 01 as the index.
- [] Formal: 12 tasks plus datapath_pdr pass, and the vacuity check catches 19/19. The code-3 branch is vacuous (K-10). The mutation campaign kills 35 of 40 mutants: m07, m20, m22 and m27 survive as test gaps, and m21 is argued equivalent. Question 10.
- [] Fault injection: the RTL SEU campaign and the gate-level campaign on the routed sep netlist (702 duplicated-copy flips stopped, 234 thermal-copy flips repaired, 0 mismatches). Known escapes: out_valid_q 200/200, fault_q 0->1 200/200 false stops, phase 44/200 shifted admissions; fault_q 1->0 is not measured. neg_product shows silent data corruption (K-03).
- [] Every simulation is zero-delay. No SDF-annotated gate-level simulation exists (MISSING). Decide whether that is needed.
- [] Waveforms in [05_simulation/waveforms/](#) are plotted from VCDs of RTL runs at a 10 ns functional clock. Spot-check scenario (c), the fault-stop, against SPEC section 5.

3.3 Timing and constraints

Documents: [07_verification/verification_summary.pdf](#) timing sections; [sta_corners_sep_2026-10-06/sta_corners_sep.txt](#); SDC [06_physical_design/layout_db/6_final.sdc](#).

- [] Constraints: one clock at 7.2 ns, input and output delays of 20 % of the period (1.44 ns, an ASSUMPTION), no load, drive, input transition, uncertainty or derate. Questions 1-2.
- [] TT: setup WNS 0.031 ns, hold WNS 0.436 ns, fmax 139.49 MHz including the I/O paths (170.82 MHz reg-to-reg). The critical path is in2reg, from in_a[27] through the lane's shared multiplier to g_lane[3].u_lane.u_res_a/q[29].
- [] ss_100C_1v60 (re-run for this package): WNS -6.131 ns, 516 endpoints, min period 13.33 ns, also 37 max-slew and 12 max-cap violations. FF passes. Only these three liberty corners exist on disk, and the SPEF has one nominal RC corner.
- [] LEC: both kepler-formal checks report "Circuits are IDENTICAL". The "4_rsz" check compares the netlist before and after repair_timing, not synthesis against post-CTS (K-14). The compared netlists are in [additional_evidence/lec_netlists/](#).

3.4 Floorplan and layout

Documents: [06_physical_design/physical_design.pdf](#); full-resolution images in [klayout_views/](#) and [orfs_images/](#).

- [] Die 346.295 um square, utilization 0.5361. The PDN has met1 rails and met4/met5 straps, single VDD/VSS.
- [] Copy-separation fences (v03, v04, v05): copy A on the left edge, copy B on the right, three thermal fences. Accumulator and result same-bit copies are at least 251.17 um apart, thermal copies 97.92 um, and none of the 262 same-bit pairs touch. Non-member cells overlap the fences, including clock-tree buffers (K-13; question 13). Shared drivers across copies (D-93; question 6).
- [] Open the GDS in KLayout and check one area yourself, for example the cells in and around a thermal fence. The fences are not drawn in the GDS: take the box coordinates from 06 section 4.1.

3.5 Physical verification, power and reliability

Documents: [07_verification/verification_summary.pdf](#).

- [] DRC: TritonRoute 0, KLayout 0 markers. The deck has FEOL disabled, no density rules and no fill, so this is not sign-off DRC (K-04; question 7).
- [] LVS: KLayout, netlists match, but the reference CDL is written from the same database as the GDS. No independent Netgen LVS.
- [] Antenna: 0 violations, 20 diodes.
- [] Power 52.1092 mW and IR drop (VDD 0.4539 mV, VSS 0.6275 mV) both use default activity at TT. They are tool estimates, not measurements (K-06; question 11).
- [] Electromigration: no check against current-density limits (MISSING); a package re-run gives grid currents only (K-06).

3.6 Pinout and packaging

Documents: [08_pinout_packaging/pinout_packaging.pdf](#); [pinout.csv](#).

- [] 217 pins = 215 signal bits of the 18 RTL ports + VDD + VSS, placed on the die edge by the ORFS pin placer.
- [] No pad ring, ESD, seal ring or package (K-05), and no electrical ratings (G-15). Check the external-connection list: clock source, reset, sensor inputs and supply.

3.7 Provenance

Documents: README revision table; [09_review_notes/review_notes.pdf](#) section 4.3.

- [] The OpenROAD version is "unknown", no PDK commit is logged, and linking the run to the ORFS image digest is an ASSUMPTION. The SDC with the 7.2 ns period was committed after the run started (K-09, D-52; question 12). Decide whether the run must be repeated from a pinned image and a clean commit.

Step 4. Revision and consistency check (15 min)

- [] Verify the package files: run `sha256sum -c MANIFEST.sha256` in the package folder. Every line should end OK.
- [] Skim the discrepancy register (review_notes section 4). Most items are inconsistencies between files; some are marked fixed in the package, and the rest are for the design owner to resolve. Flag any you consider blocking.
- [] Skim the gaps table (review_notes section 5). It lists what a complete review needs that does not exist.

Step 5. Reproduce selected results (optional)

From a repository checkout at the package commit (see [04_digital_design/README.md](#) section 6.1 for setup). Times are the 2026-10-06 package re-runs (04 sections 6.3 and 6.4).

What	Command	Expected	Time
Lint	<code>make lint</code>	exit 0, no %Warning lines	0.12 s
Directed simulation	<code>make sim-directed BUILD=/tmp/r</code>	TB_ORBIT_DEMO PASS	6.5 s
Formal	<code>make formal BUILD=/tmp/r</code>	line starting [formal] all 12 tasks passed	47 s
Gate-level fault campaign, routed sep netlist	<code>make pd-sep-gls PDSEP_PUBLISH= (needs build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.v from a make pd-sep run, and Docker for the cell library)</code>	GLS PASS: 30000 cycles, 0 mismatches	2 min 18 s (the target's commands run by hand)
Multi-corner STA of the sep layout	<code>run_corners_sep.sh</code> (edit its REPO= and OUT= paths first; needs Docker, the pd-sep outputs and the SS/FF liberty files that make pd-corners downloads to build/pd/platform/sky130hd/corners/)	one sta_<corner>.log per corner, with the values in sta_corners_sep.txt	about 2 s per corner

Use `BUILD=<dir>` and the `*_PUBLISH=` overrides so the tracked reports/ folder is not rewritten (04 section 6.2).

Step 6. Write feedback

Reference the package IDs so each comment can be traced. Questions are 1-14, known issues K-01..K-16, trade-offs T-01..T-07, discrepancies D-01..D-101 and gaps G-01..G-38, all in the review notes. Drawing items BD-1..BD-5 and SC-1..SC-4 are in 02_block_diagram and 03_schematics. Template:

#	Ref (Q / K / T / D / G / file:section)	Finding or answer	Severity (blocking / major / minor / note)	Requested action
1	Q1			
2	K-01			
3				

Close with an overall outcome, for example: ready for the next step; ready with listed conditions; or not ready, with the blocking items. Say which step the outcome applies to, for example "continue design iteration" or "start pad-ring and sign-off DRC/LVS work". The package does not claim tapeout readiness.